STATISTICS WORKSHEET-4

1) The CLT is a statistical theory that states that - if you take a sufficiently large sample size from a population with a finite level of variance, the mean of all samples from that population will be roughly equal to the population mean.

   It is important because-

   - The data must adhere to the randomization rule. It needs to be sampled at random.
   - The samples should be unrelated to one another. One sample should not impact the others.
   - When taking samples without replacement, the sample size should not exceed 10% of the population.

When the population is symmetric, a sample size of 30 is generally considered reasonable.

2) The number of elements or individuals in a sample is called the sample size. The process of selecting a sample is called **sampling.** Sampling techniques

   Simple Random Sampling: Every member of this sample has an equal chance (probability) of selection is called Simple Random Sampling

   Systematic Sampling : In this sampling technique, we systematically select members. In particular, members are chosen at regular intervals of the population by putting all the members in a sequence first.

   Cluster Sampling **:** Cluster sampling is often confused with stratified sampling but both these sampling techniques are different from each other. The main difference is that with

cluster sampling you have natural groups separating your population. For example, clusters like city blocks, school districts, age, sex, etc.

3) In statistics, a Type I error is a false positive conclusion, while a Type II error is a false negative conclusion.

Making a statistical decision always involves uncertainties, so the risks of making these errors are unavoidable in hypothesis testing.

The probability of making a Type I error is the significance level, or alpha (α), while the probability of making a Type II error is beta (β). These risks can be minimized through careful planning in your study design.

Example: Type I vs Type II error You decide to get tested for COVID-19 based on mild symptoms. There are two errors that could potentially occur:

- Type I error (false positive): the test result says you have coronavirus, but you actually don't.
- Type II error (false negative): the test result says you don't have coronavirus, but you actually do.

4) Normal distribution describes how the values of a variable are distributed. It is the most important probability distribution in statistics because it accurately describes the distribution of values for many natural phenomena.

5) Covariance is a statistical term that refers to a systematic relationship between two random variables in which a change in the other reflects a change in one variable.

In statistics, correlation is a measure that determines the degree to which two or more random variables move in sequence. When an equivalent movement of another variable reciprocates the movement of one variable in some way or another during the study of two variables, the variables are said to be correlated

6) Univariate Data Analysis

Being the underlying form of data analysis, univariate has just one variable in its analysis. Hence, it lacks any derived study like causal relationships ,To buttress, this makes use of a single variable like the amount of clients who purchased an online course in a bid to study these numbers to bring about useful insights while calculating their dispersion, mean, median, mode, standard deviation, range or variance and visualizing its results in form of a bar or pie chart, histogram, table, frequency polygons, etc.

Bivariate Data Analysis

Secondly, bivariate analysis deals with the evaluation of a pair of variables together in order to establish a kind of relationship or correlation Such variables may either be dependent or independent in their X and Y values. This research can be done using regression analysis, scatter plot, one-way ANOVA, Pearson's Chi-square, Fisher's Exact, McNemar's Test, Mantel[1]Haenszel, Spearman's Correlation, Simple Linear Regression, etc.

Multivariate Data Analysis

Thirdly, multivariate data contains over 2 variables – that is in their threes, fours, fives and so on. Here, analysis focuses on more than one variable at once. Some methods of this analysis are Analysis of Variance, Analysis of Covariance, Repeated Measures, Factor Analysis, Logistic Regression, Discriminant Analysis, multivariate analysis of variance (MANOVA), etc.

7) The technique used to determine how independent variable values will impact a particular dependent variable under a given set of assumptions is defined as **sensitivity.**

The sensitivity is calculated by dividing the percentage change in output by the percentage change in input.

8) Hypothesis Testing is an assertion or conjecture concerning one or more populations. To prove that a hypothesis is true, or false, with absolute certainty, we would need absolute knowledge. That is, we would have to examine the entire population. Instead, hypothesis

testing concerns on how to use a random sample to judge if it is evidence that supports or not the hypothesis.

Hypothesis testing is formulated in terms of two hypotheses: • H0: the null hypothesis; • H1: the alternate hypothesis

Null Hypothesis(H0) : Null hypothesis is a statement about the population or the whole dataset, that we try to reject with the sample data. It is a statement about the whole dataset or population that is either believed to be true or can be used to put forth an argument unless it is proved to be false.

Alternate Hypothesis(H1): This hypothesis is contradictory to what we try to state about the population in null hypothesis.

9) Quantitative data is anything that can be counted or measured; it refers to numerical data. Qualitative data is descriptive, referring to things that can be observed but not measured—such as colors or emotions.

10) The range is calculated by subtracting the lowest value from the highest value.

The IQR describes the middle 50% of values when ordered from lowest to highest.

11) Bell curve distribution is probability distribution that is symmetric about the mean, showing that data near the mean are more frequent in occurrence than data far from the mean.

12) . Data visualization method.

- Statistical tests (z scores)
- Interquartile range method.

13) The p value is a number, calculated from a statistical test, that describes how likely you are to have found a particular set of observations if the null hypothesis were true. P values are used in hypothesis testing to help decide whether to reject the null hypothesis.

14) The binomial distribution formula is used in statistics to find the probability of the specific outcome-success or failure in a discrete distribution.

15) ANOVA- Analysis of Variance

ANOVA is to test for differences among the means of the population by examining the amount of variation within each sample, relative to the amount of variation between the samples. Analyzing variance tests the hypothesis that the means of two or more populations are equal.